

Troubleshooting

Objectives

After completing this lesson, you should be able to:

- Describe the two-phased approach to troubleshooting
- Describe the type of information needed to troubleshoot a problem
- Describe the available operating system logs to assist in troubleshooting
- Use the `dmesg` utility
- Describe the available troubleshooting resources
- Describe causes of common problems
- Describe troubleshooting boot problems
- Describe typical causes of NFS problems



Two-Phased Approach to Troubleshooting

- Fault Analysis Phase
 - State the problem.
 - Gather information.
 - Identify what is and what is not working.
- Fault Diagnosis Phase
 - Based on the fault analysis findings and past experiences, determine the most probable causes of the fault.
 - Test and verify the probable causes.
 - Take corrective action.
 - Ensure you do not introduce any new problems.

Document the results of the fault analysis and fault diagnosis phases.

Gathering Information

- Get a complete description of the server.
- Describe exactly what the problem is.
 - Symptoms
 - Error messages
- Who is experiencing the problem?
 - One user or several users
- Can the problem be reproduced?
 - Steps to reproduce the problem
 - Is it an intermittent problem?
- Does the problem occur only at certain times of the day or certain days of the week?
- Have any changes been made to the server?



Operating System Logs

- Files under `/var/log`:
 - `boot.log` – Messages from bootup
 - `messages` – Standard system error messages
 - `anaconda` – O/S install logs
 - `dmesg` – Log of boot messages showing hardware errors
 - Other logs exist for mail, `cron`, security, and so on.
 - Other directories in `/var/log/` exist for cups, httpd, samba, and so on.
- You can monitor a log file in real time by using the following command:

```
# tail -f <logfile>
```

- To view the journal in live view:

```
# journalctl -f
```



dmesg Utility

- `dmesg`: Print out a buffer showing latest hardware issues.
- The command prints only a memory structure (kernel ring buffer) in the memory.
- `dmesg` can display human readable time stamps with the “-T” option:
 - `dmesg -T`
- The buffer can truncate when it is full.
 - `/var/log/boot*`
 - `/var/log/dmesg*`

Troubleshooting Resources

- Man pages provide the usage of a command and the available options and configuration parameters.
- Many commands and services have a `-d/-D` option for debugging or a `-v/-V` option for verbose.
- The `/usr/share/doc` directory contains information about packages installed on your system plus release notes and manuals.
- Oracle Linux 7 administration guides:
 - http://docs.oracle.com/cd/E52668_01/
- The My Oracle Support website contains knowledge articles and other helpful information.
 - <https://support.oracle.com/>



My Oracle Support

The screenshot shows the My Oracle Support interface. At the top, there's a navigation bar with tabs: Dashboard, Knowledge (selected), Service Requests, Patches & Updates, Community, Certifications, Managed Cloud, CRM On Demand, and More... A search bar is on the right. Below the navigation bar, the 'KM Search Results' section is active. It shows filters for 'Knowledge Base', 'Archive', and 'Documentation'. The search query 'linux boot' is entered. Below the search bar, there are filters for 'Start typing to select Product', 'Any Product Version', 'Any Platform', and 'Last updated - Any time'. The results are displayed under the heading 'Knowledge Base Search Results'. There are eight results listed, each with a star icon, a document icon, a date, a title, and a document ID. The first result is 'How to Boot Linux OS to Rescue Mode with Multipathing' (Doc ID 2302834.1) dated Sep 11, 2017. The second is 'Exalogic Compute Node Stuck At GRUB Linux Boot Sequence' (Doc ID 1370639.1) dated Jul 23, 2012. The third is 'How To Boot Linux Paravirtualized VM Into Rescue Mode and in Single User Mode' (Doc ID 1602157.1) dated Sep 24, 2017. The fourth is 'How to boot Oracle Linux in Interactive Mode' (Doc ID 1500952.1) dated Nov 19, 2012. The fifth is 'ILOM Java console appeared as Linux System in Hung during boot.' (Doc ID 2190274.1) dated Nov 24, 2016. The sixth is 'How to Boot Oracle Linux 5.x into Rescue Mode' (Doc ID 1516777.1) dated Jan 29, 2016. The seventh is 'Quickest and easiest way to boot a PVM Linux guest into Rescue Mode' (Doc ID 2187003.1) dated Sep 14, 2017. The eighth is 'How to Configure an Oracle Linux PXE Boot Server' (Doc ID 578582.1) dated Aug 25, 2013.

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Results: Knowledge Base

Tips
Use the "inline" product refinement within your search result lists to quiddly broaden or narrow down your search results.

Knowledge Base Search Results

- ★ Sep 11, 2017 **How to Boot Linux OS to Rescue Mode with Multipathing** (Doc ID 2302834.1)
Refine to All > Oracle Linux and Virtualization > Oracle VM > Oracle VM > Oracle VM
- ★ Jul 23, 2012 **Exalogic Compute Node Stuck At GRUB Linux Boot Sequence** (Doc ID 1370639.1)
Refine to All > Middleware > Exalogic > Exalogic Software > Oracle Exalogic Elastic Cloud Software
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Causes of Common Problems

- Service(s) not running:
 - Use the `systemctl` command to start a service or check the status of a service.
 - Use the `systemctl enable` command to start a service at boot time.
- Configuration errors:
- Firewall (`firewalld` and `iptables`) is prohibiting a connection.
 - Stop the service and test to determine if a firewall is blocking.
- PAM is prohibiting authentication:
 - View `/var/log/secure` for authentication error messages.
- SELinux is denying a connection:
 - Set SELinux to permissive mode and test.

Troubleshooting Boot Problems

- Configuration errors in the following files can prevent your system from booting:
 - `/boot/grub2/grub.cfg`
 - `/etc/fstab`
- Boot into rescue mode to correct boot problems.
 - Rescue mode boots from installation media.
 - File systems are mounted under `/mnt/sysimage`.
 - Use `chroot` to change the root partition of the rescue mode environment.
 - Then use `vi`, `fsck`, `rpm`, and other utilities to fix the boot problem.
- Use the `grub2-install` to re-install the boot loader.



Typical Causes of NFS Problems

- The `rpcbind` or NFS daemons are not running:
 - NFS daemons are `nfs` and `nfslock`.
- Syntax errors:
 - On client `mount` command
 - In `/etc/exports` file on server
- Permission problems:
 - Check UIDs and GIDs.
- Firewall is blocking NFS packets:
 - Check `firewalld` and `iptables` rules or stop the service.
- DNS host name resolution:
 - Ensure `/etc/resolv.conf` contains correct entries.

Quiz



Which command is useful in determining if your system has hardware-related errors?

- a. `service`
- b. `ps`
- c. `lsmod`
- d. `dmesg`



Summary

In this lesson, you should have learned about:

- Two-phased approach to troubleshooting
- Type of information needed to troubleshoot a problem
- Available operating system logs to assist in troubleshooting
- Use of the `dmesg` utility
- Available troubleshooting resources
- Causes of common problems
- Troubleshooting boot problems
- Typical causes of NFS problems



Practices for Lesson 20: Overview

The practices for this lesson involve troubleshooting the following common problems:

- System boots into single-user mode
- Status commands fail
- A `cron` job fails to run
- A user cannot log in
- File system does not mount
- Network connectivity problem
- You cannot log in to remote hosts using `ssh`
- Log file is not getting updated